

Mitsue Project × NAIST CDG — One-Page Overview

Prepared for Prof. Minoru Kubo, Center for Digital Green-innovation · 2026-07-28

Biomass Energy & AI — Reforesting in Mitsue is a 25-year initiative in Mitsue Village (御杖村, Nara) that closes the loop between forest, energy, heat, and computation at village scale.

The loop

- **Forest** — Convert neglected sugi monoculture to native broadleaf. Restores ecology, feeds wildlife (keeping deer/boar in the forest), and the thinnings become fuel.
- **Energy** — Biomass CHP from sugi thinnings gives dispatchable 24/7 electricity + heat (primary); solar + battery are a secondary hedge.
- **Compute** — A small, efficient AI data center powered by that local energy. Compute is the CHP's best customer — worth far more per kWh than grid export — which is what makes the energy investment viable.

The three pillars reinforce each other; none works as well alone. The work is carried out by the village forestry cooperative (地域林業パートナー), which the project aims to roughly double over time.

Why this is interesting to CDG

- **A living lab** — a real off-campus site to study green-powered compute and heat reuse, alongside NAIST's satoyama and regional-SDGs work. Open environmental, energy, and forestry data (an ethos inherited from Safecast, 250M+ open measurements since 2011).
- **NAIST SDGs Action** — our native-broadleaf replanting and field data collection can host a student program: real trees, sensors, and data in a working village. We already run a bilingual field app (GPS + photo, per-volunteer login) built for exactly this.

Status

Phase 0 — feasibility, community consultation, NPO formation. No major commitments made; no funding sought from NAIST. This is an exploration of genuine fit.

Who

Rob Oudendijk — Dutch electrical engineer, Mitsue resident since 2012, core hardware developer for [Safecast](#). Intro via NAIST President Kaz Shiozaki.

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